

Valmet DNA DCS – Exam Preparation Summary

1. DCS vs. PLC

- **DCS (Distributed Control System)** – designed for **large, continuously operated plants** (e.g., power plants, refineries, pulp & paper).
- **PLC (Programmable Logic Controller)** – suited for **machine-level or discrete automation** (single machines, packaging lines).
- **Key differences**
 - **Scale & I/O:** DCS handles thousands of I/O points; PLC typically fewer.
 - **Operator interaction:** DCS requires continuous operator supervision; PLC can run autonomously.
 - **Architecture:** DCS distributes control functions across multiple stations/nodes; PLC often centered on one or a few controllers.
 - **Lifecycle:** DCS has long lifecycles with gradual upgrades; PLC upgrades are usually more abrupt.

2. System Diagram & Architecture

- **Distributed topology** – multiple **stations** (process, backup, alarm, engineering) connected via **redundant Ethernet** or fieldbus (PROFINET, Profibus, OPC-UA, Modbus).
- **Core components**
 - **Process Station (PCS)** – executes control logic, communicates with field devices.
 - **Backup Station (BU)** – distributes software, hosts applications, provides automatic recovery.
 - **Alarm Station (ALS/ALP)** – centralizes alarm collection, hierarchy, and distribution to operator stations.
 - **Engineering Activity Server (EAS)** – hosts FbCAD/SeqCAD diagrams, manages versions, provides function-test and graphical design tools.
- **Redundancy** – stations use redundant communication paths for high availability.

3. Station Naming & Roles

Part of name	Meaning	Example
Subsystem letter	Plant area / functional group	G = Automation lab, E = Energy technology lab
Station type code	Function of the station	P = Process Controller, A = Alarm, B = Backup, E = Engineering Activity Server, O = Operator, D = Diagnostic, OPC = OPC station
Station number	Unique instance within the subsystem	01, 10, 09

Typical examples

- **GP01** – Subsystem G, Process Controller, station 01.

- **GB01** – Subsystem G, Backup station, number 01.
- **EAS-G** – Engineering Activity Server in Subsystem G.

4. I/O Address Allocation

- **Tag naming**: must be **globally unique**; use the **pr:** prefix (e.g., **pr:MITTAUS:B1**).
- **Direct Access ports** – global identifiers, must be unique system-wide (e.g., **pr:MITTAUS:F:out1**).
- **Interface ports** – can be reused inside a single diagram but must be unique **per diagram** (e.g., **:in1**, **:out1**).
- **Data types**
 - **Analog** – float values, green wire (e.g., **pr:xxxx-TT-100.I**).
 - **Binary** – bit values, black wire (digital input/output).
 - **Structured** – composite signals (fault bits + value).
- **Signal manipulation** – use **ccob**, **ccox** blocks with **condition (cnd)** and **mode** parameters for edge detection, filtering, or conditional copying.

5. Function Test Tool

- **Purpose** – verify correctness of a function-block or sequence diagram by comparing live/runtime values with expected results.
- **Typical workflow**
 1. **Start** the tool (File → Test) from DNA Explorer, FbCAD, or SeqCAD.
 2. **Add test points** (toall, toarea, toselected) – they appear as blue numbers linked to connection points.
 3. **Enter values** via the test-point dialog; the diagram updates instantly.
 4. **Check responses** – observe if other test points change as expected.
 5. **Use History view** – pause, step forward/backward, or jump to a specific cycle to replay past behavior.
 6. **Create trends** – insert trend areas to visualise value changes over time.
- **Important notes**
 - Test on the **Repository** workspace, not directly on the process station.
 - Ensure **Real** mode (not Virtual) when real I/O is involved.
 - After testing, **remove** the loop from the process station and **delete** it from the repository to avoid IO address conflicts.

6. Key Concepts to Remember

#	Concept	Core Idea
1	Same fundamental control tasks	Both DCS and PLC read inputs, compute control, write outputs.
2	Integrated plant-wide automation	DCS provides a coordinated, scalable environment across the whole plant.
3	Continuous operator interaction	Operators monitor, intervene, and manage alarms in real time.

#	Concept	Core Idea
4	Distributed specialised stations	Functions are split among dedicated stations, enabling easier scaling and maintenance.
5	Long lifecycle & gradual modernisation	Systems evolve over years; upgrades are incremental rather than replacement-only.

7. Example Exam Questions & Answers

Question	Answer
What is a key feature of DCS systems compared to PLCs?	DCS provides continuous operator interaction and distributed, plant-wide control , while PLCs are typically used for smaller, machine-level automation with less direct operator involvement.
What does DCS stand for?	Distributed Control System
What does GB01 indicate in naming policy?	G = Subsystem, B = Backup station, 01 = station number.
What is the role of the backup station in Valmet DNA?	It distributes software, hosts applications, provides automatic recovery after power failures, and centralises management of software and configuration across all nodes.
What does the black wire colour represent in the Valmet DNA software?	The black wire denotes a binary (digital) signal – either 0 or 1 – used for digital inputs or outputs.
What happens if two blocks have the same tag name in Valmet DNA?	The second block overwrites the first ; tags must be unique to avoid conflicts and ensure correct data handling.
What does the Function Test Tool allow you to do?	It lets you test diagrams with live/runtime values, add test points, enter values, step through historical data, and visualise trends to verify correct functionality.

8. Additional Possible Exam Questions (in the same style)

Question	Expected Answer
What are the main components of a Valmet DNA DCS and their primary functions?	Process Station (PCS) executes control logic; Backup Station (BU) distributes software and provides recovery; Alarm Station (ALS) manages and distributes alarms; Engineering Activity Server (EAS) hosts engineering tools and manages application lifecycle.

Question	Expected Answer
Explain the naming convention for Valmet DNA stations. What does each part signify?	The name consists of a subsystem letter (identifies plant area), a station type code (indicates function: P=PCS, A=Alarm, B=Backup, E=EAS, O=Operator, D=Diagnostic, OPC=OPC station), and a station number (unique instance within subsystem).
How does the Function Test tool help in verifying a control loop?	It allows live values from the runtime environment to be displayed as test points on the diagram, enables entering values to stimulate the loop, and provides history and trend views to analyze behavior over time.
What is the purpose of the Backup Station in Valmet DNA, and how does it contribute to system reliability?	The Backup Station distributes software to all nodes during download, hosts applications for retrieval, enables automatic recovery after power failure, and provides centralized management of software versions, ensuring consistency and reducing downtime.
Describe the difference between a Direct Access Port and an Interface Port in Valmet DNA FbCAD.	A Direct Access Port is a global identifier (must be unique system-wide) used to share data across the entire automation system, while an Interface Port is local to a diagram (must be unique within that diagram) used to share signals between programs within the same diagram.
What data type is represented by a black wire in Valmet DNA, and what values can it take?	A black wire represents a binary data type, which can take values 0 or 1 (off/on).
What is the role of the Engineering Activity Server (EAS) in the Valmet DNA lifecycle?	The EAS hosts the engineering database (FbCAD/SeqCAD diagrams), provides graphical design tools, manages version control, supports commissioning and maintenance, and includes the Function Test tool for verifying applications before deployment.
How does Valmet DNA achieve scalability, and what remains coordinated despite distribution?	Valmet DNA scales by adding stations, nodes, or process areas; despite distribution, the system remains a one coordinated automation environment with shared alarms, trends, process information, and a unified engineering environment.
What are the typical fieldbus protocols supported by Valmet DNA ACN I/O?	Valmet DNA supports ACN, PROFINET, Profibus, OPC/OPC UA, Modbus (RTU/TCP), and others via ACN I/O modules.
In the context of I/O allocation, why must tag names be unique in Valmet DNA?	Tag names are used as global identifiers in the database; duplicate names would cause conflicts, overwriting data and leading to incorrect system behavior or download failures.
What is the purpose of the ccob (conditional copy) block, and how are its behavior controlled?	The ccob block copies a signal when a condition is met; its behavior is controlled by the condition (cnd) and mode

Question	Expected Answer
	parameters, enabling functions like edge detection, filtering, or gating signals.
What happens if you attempt to download an application with a tag name that already exists in the system?	The download will fail or overwrite the existing application, causing conflicts; you must use a unique tag name or rename the existing one before downloading.
What is the difference between a Process Station (PCS) and a Backup Station (BU) in terms of operating system and primary function?	The PCS typically runs a modified Linux OS for real-time control, while the BU runs Windows OS (or can be virtualized) and focuses on software distribution, application hosting, and system recovery.
How are alarms managed in Valmet DNA, and what role does the Alarm Station play?	Alarms are collected from various components, timestamped, categorized, and prioritized by the Alarm Station, which then distributes them to operator stations for awareness and action.
What is the significance of the pr: prefix in Valmet DNA tag names?	The pr: prefix indicates that the tag is a global identifier (position name) used throughout the automation system, ensuring uniqueness across all nodes.
What is the role of the DNA Explorer in the Valmet DNA engineering workflow?	DNA Explorer is the main user interface for managing application programs, allowing users to download, upload, and organize applications, as well as access tools like Function Test and viewers for diagrams.
What are the typical I/O modules available in Valmet DNA ACN I/O, and how many channels do they provide?	Typical modules include DI8 (8 digital inputs), DO8 (8 digital outputs), AI8 (8 analog inputs), AO8 (8 analog outputs), and TI4 (4 temperature inputs), with 16 cards fitting in one ACN I/O unit.
What is the purpose of the IO Check function in Valmet DNA FbCAD?	The IO Check function validates the I/O configuration of an application, ensuring that all modules are correctly defined and that there are no conflicts before downloading.
How does the system handle a power failure, and which component enables automatic recovery?	The Backup Station (BU) stores applications and configuration; after a power failure, nodes retrieve their software and settings from the BU, allowing the system to restart automatically without manual intervention.
What is the difference between a wired I/O connection (as in traditional wiring) and a fieldbus connection in Valmet DNA?	Wired I/O uses individual cables from field devices to I/O modules (often via trunk and cross-connection cables), while fieldbus (e.g., Profibus, PROFINET) uses a shared communication bus to connect multiple devices with fewer cables.
What is the role of the Cross connection in traditional I/O wiring for Valmet DNA?	The cross connection (often using JAMAK/NOMAK terminology) is a terminal block in the I/O cabinet where field device cables

Question	Expected Answer
	are terminated and connected to trunk cables leading to the I/O modules, providing a flexible and organized wiring structure.

9. Additional Topics (Brief Overview)

- **Hardware Versions:**
 - ACN MR (Modular Rail) – small form factor, deprecated.
 - ACN CS (Compact Server) – industrial PC, deprecated.
 - ACN RT (Rack Mount) – most powerful, current option.
- **Fieldbus Support:** ACN, PROFINET, Profibus, OPC-UA, Modbus, etc., via ACN I/O modules.
- **Scalability:** Add stations, nodes, or process areas; system remains one coordinated automation environment.
- **Alarms & Trends:** Centralized alarm management with hierarchy and timestamping; trend analysis for process monitoring.
- **Engineering Lifecycle:** Design → Test (Function Test) → Download → Operate → Maintain → Upgrade, supported by EAS tools.
- **Safety:** Integrated Safety Instrumented System (SIS) options with SIL levels.
- **Operator Interface:** Valmet DNA Operate provides intuitive tools for operation, alarm handling, trends, reporting, and history.

This summary is intended for exam preparation. Review all provided materials, especially the documents and videos listed in the directory, for full understanding.